AMERICAN BIOCARBON
100% PREMIUM BIOCHAR - TECHNICAL SPECIFICATION SHEET

THE ONLY US-PRODUCED SUGARCANE BIOCHAR

Most biochar sold in the United States is made from wood. American Biocarbon makes biochar from sugarcane bagasse, which gives it a different structure, a different mineral profile, and a different job to do in the soil.

Sugarcane bagasse, the fibrous byproduct of cane sugar production, gives our biochar a uniform, continuously channeled microstructure that holds more water, retains more nutrients, and supports richer microbial habitat than wood-based alternatives.

American Biocarbon is the only US producer of sugarcane biochar sold as a packaged product, manufactured at scale alongside the Cora Texas sugar mill in White Castle, Louisiana.

CORE SPECIFICATIONS (DRY BASIS)

Organic Carbon:	~58–65%
Ash Content:	~25–40%
H/C Molar Ratio:	0.38–0.61 (IBI Class 1 stability, ≤ 0.7)
pH:	7.6–9.3
Surface Area:	215–300 m^2/g
Bulk Density:	10–23 lb/ft^3
Electrical Conductivity:	<0.35 dS/m
Moisture Content:	24–52% (wet basis, batch-dependent)
Volume Yield:	~4 cubic yards/MT (wet)
Form:	Granular
Odor:	Low, earthy

INHERENT NUTRIENTS

Nitrogen (N):	0.64%
Phosphorus (P):	0.08%
Potassium (K):	0.55%
Calcium (Ca):	0.8–1.5%
Magnesium (Mg):	1.0–2.0%

Heavy Metals: 1–2 orders of magnitude below IBI and EPA Class A thresholds

THREE-PILLAR VALUE PROPOSITION

STABLE MICROSTRUCTURE

The vascular architecture of sugarcane bagasse survives pyrolysis intact. SEM imaging shows uniform, continuously channeled pores at 10–50 μm — the kind of consistent internal geometry that holds water, retains nutrients, and gives beneficial microbes a protected habitat.

INHERENT MINERALS
Sugarcane biochar delivers inherent N-P-K (roughly 0.6 / 0.2 / 0.7 as fertilizer equivalent) along with calcium, magnesium, and trace minerals retained from the cane. These nutrients release gradually through soil's natural processes.

VERIFIED CONSISTENCY
Independent analyses across multiple production batches show heavy metals run one to two orders of magnitude below IBI and EPA Class A thresholds. H/C molar ratio consistently within the IBI Class 1 stability threshold. OMRI Listed for organic crop production.

HOW IT COMPARES TO WOOD BIOCHAR

Property	Wood Biochar	American Biocarbon Bagasse
Pore geometry	Random, fractured	Ordered honeycomb
Pore connectivity	Low to moderate	High
Plant-available water capacity	Strong	Strong, with more uniform release
Nutrient retention mechanism	Adsorption only	Adsorption + inherent N-P-K
Mineral / ash content	Low	Moderate (with usable Ca, Mg, K)
Microbial habitat	Limited protected niches	Continuous protected channels
Carbon content by weight	Higher (typically 70–90%)	Moderate (58–65%)

HONEYCOMB MICROSTRUCTURE: THE SUGARCANE ADVANTAGE

The pore geometry difference is significant:
- Wood biochar: Irregular, fractured pore network with collapsed vessels and disconnected channels
- American Biocarbon bagasse biochar: Uniform honeycomb architecture with continuous, interconnected channels at 10–50 μm

This structural difference means:
- Faster water infiltration and more uniform distribution
- Better nutrient retention with more consistent release
- Enhanced microbial habitat throughout the material
- Superior performance in sandy soils and salt-affected soils
- More efficient use in compost and blending applications

APPLICATION NOTES

Bagasse biochar performs particularly well in:

- Sandy soils – improves water retention and nutrient availability
- Salt-affected soils – structural improvement with mineral support
- Degraded ground – where structure, water retention, and fertility need to be rebuilt together
- Heavy clay – improved drainage and aeration through continuous pore network
- Row crops – specialty crops, horticulture, land reclamation, and remediation work where the combined structural-and-mineral profile does the job of two separate amendments

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CERTIFICATIONS & COMPLIANCE

✓ OMRI Listed – Organic input listing for organic crop production
✓ IBI Certified – International Biochar Initiative program verification
✓ Puro.earth Certified – Carbon removal verification (where applicable)
✓ Independent Lab Analysis – Heavy metals well below IBI / EPA Class A thresholds
✓ USDA-ARS Field Study – Multi-year sugarcane bagasse biochar trials

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PACKAGING OPTIONS

- Super sacks: 3, 3.6, or 4 cubic yards
- Bulk shipment available for large orders

CONTACT & ORDERING

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